

JONATHAN HE

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SUMMARY

Software engineer specializing in identity, authorization, and AI security infrastructure. Architected delegated authorization, cross-domain identity propagation, and agent security controls for Amazon's foundational authorization platform supporting 100K+ applications. Experienced leading security systems from technical design through production rollout and customer adoption.

EXPERIENCE

Amazon

New York, NY

Software Development Engineer II

Dec 2024 – Present

- Architected Amazon's delegated authorization model, enabling AI agents and services to act on behalf of users while preserving least privilege access across a platform supporting 100K+ applications.
- Delivered end-to-end on-behalf-of authorization for agentic workflows, preserving identity and authorization context across service boundaries.
- Prototyped and integrated an agent security enforcement layer that intercepts tool calls, applies centralized policies, routes ambiguous actions to an LLM-as-judge, and records activity for audit and detection.
- Led AWS IAM Identity Center Trusted Identity Propagation interoperability, enabling enterprise identities to propagate securely across internal and AWS trust domains.
- Designed highly available public distribution infrastructure for OIDC metadata and signing keys, enabling external systems to validate identities issued by Amazon's internal authentication infrastructure.

Software Development Engineer I

Jan 2023 – Dec 2024

- Built a distributed cache for Amazon's transaction token issuer cutting authorization latency by 80% while serving 1M+ reads per second.
- Doubled per-host capacity from approximately 8K to 16K TPS by profiling production code and eliminating a CPU bottleneck in metrics processing.
- Led stress and chaos testing across critical dependencies, exposing failure modes and redesigning load shedding to improve service resilience under overload by 30%.
- Enabled zero-downtime migration from RSA to ECDSA certificates through event-driven certificate reloading, improving cryptographic agility without interrupting authentication traffic.

Software Development Engineer Intern

May 2022 – Aug 2022

- Automated configuration and feature rollout for Amazon's authorization infrastructure, reducing deployment time by more than 48 hours and adding observability and rollback controls.

TECHNICAL LEADERSHIP

Speaker, "Agentic Authorization: Agents Need Guardrails Too" — Amazon Internal Security Conference, 50+ attendees. Presented delegated identity, least privilege, and fine-grained policy enforcement for AI agents.

EDUCATION

University of Illinois Urbana-Champaign

Dec 2022

B.S. Computer Science & Statistics, Minor in Econometrics | Cum Laude | GPA: 3.95/4.0

EXPERTISE

Security: Identity & Access Management, Authorization, OAuth 2.0, OpenID Connect, Delegated Authorization, Trusted Identity Propagation, Fine-Grained Access Control, AI Security

Programming & Infrastructure: Java, Python, TypeScript, SQL, AWS, Distributed Systems, CloudFront, Redis, DynamoDB, AWS CDK